CASE STUDY REPORT

Comparative Analysis of Classification Algorithms for Bank Marketing Prediction

Machine Learning / Artificial Intelligence

Submitted by

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1. Problem Statement

Banks use telemarketing campaigns to offer term-deposit products to customers. Contacting every customer can consume staff time and campaign resources, while failing to identify interested customers can reduce campaign effectiveness. This case study compares four classification algorithms to predict whether a customer will subscribe to a term deposit using customer and campaign information available before a call. The target variable is **y**, with **yes** indicating subscription and **no** indicating no subscription.

Objectives

• Inspect and summarize the bank marketing dataset.
• Explore subscription patterns using statistical summaries and visualizations.
• Preprocess numeric and categorical predictors and create a consistent train-test split.
• Train Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Random Forest.
• Compare models using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC.

2. Methodology

Dataset and initial inspection

The supplied *bank-full.csv* file contains **45,211 records and 17 columns**. It was loaded with a semicolon delimiter. The dataset includes numeric fields such as age, balance, day, campaign, pdays, and previous, as well as categorical fields such as job, marital status, education, contact, and previous campaign outcome.

Data quality

The initial inspection found **0 null/missing values** and **0 duplicate rows**. Some categorical fields use the literal value *unknown*; this is a category label rather than a null value. Counts of unknown entries are job: 288, education: 1,857, contact: 13,020, and poutcome: 36,959. These entries were retained as categories and handled by one-hot encoding.

Preprocessing and split

The target was mapped to 0 for *no* and 1 for *yes*. The *duration* variable was excluded because it records call duration and would not be available before the call. Numerical features were standardized, and categorical features were one-hot encoded with unknown categories handled safely. An 80:20 stratified train-test split was used with random_state=42. Preprocessing was fitted within model pipelines to avoid fitting transformations on the test set.

3. Statistical Findings

Target distribution

Of 45,211 customers, 39,922 (88.30%) did not subscribe, while 5,289 (11.70%) subscribed. The target is therefore imbalanced: the no-subscription class is much more common. This matters because a classifier may obtain high overall accuracy by correctly predicting the majority class while still missing many subscribers.

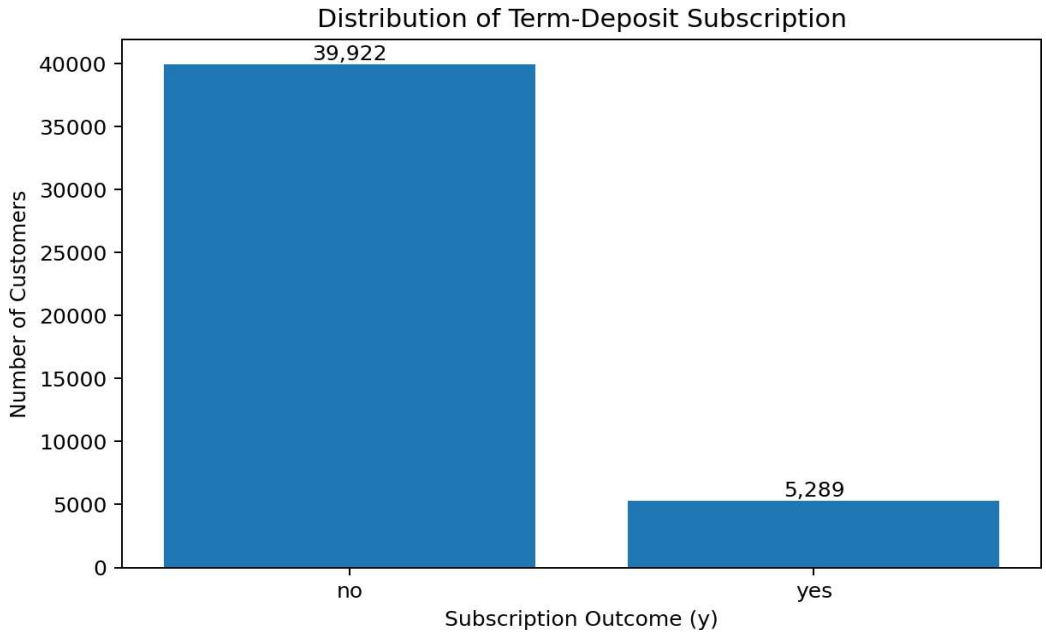


Figure 1. Subscription counts. The chart shows the strong predominance of non-subscribers.

Subscription rate by job

The observed subscription rate varied by job category. Students had the highest rate (28.68%), followed by retired customers (22.79%). Blue-collar customers had a lower rate (7.27%). These are descriptive associations in this dataset and do not establish that occupation alone causes subscription.

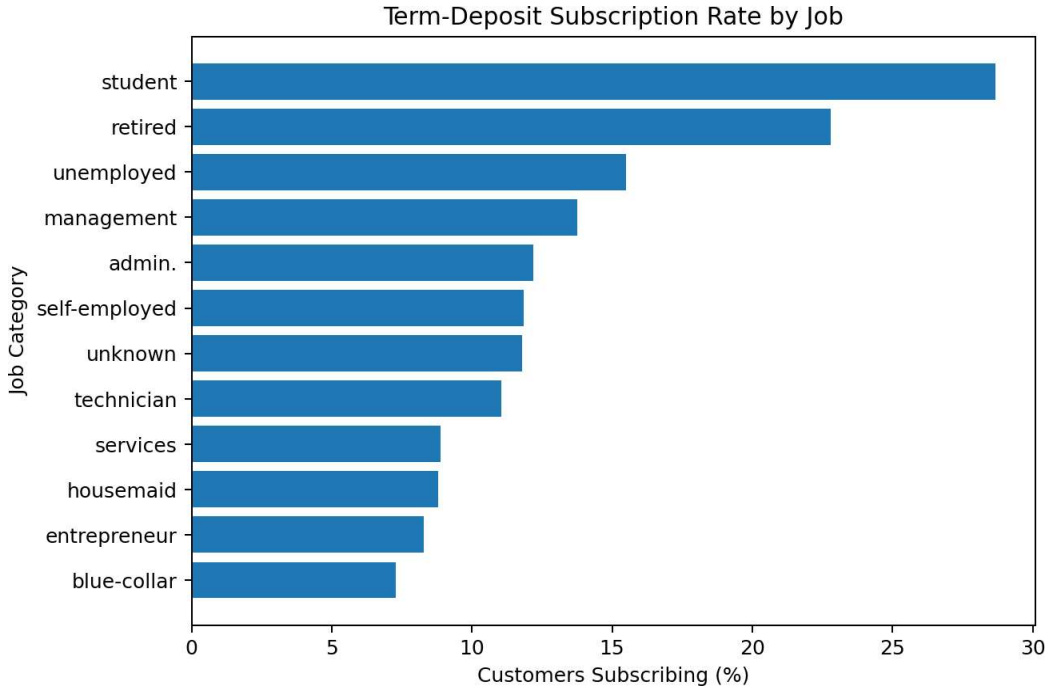


Figure 2. Percentage subscribing within each job category. Rates differ across categories.

Previous campaign outcome

Customers whose previous campaign outcome was recorded as success had a subscription rate of 64.73%, compared with 12.61% for those whose previous outcome was failure and 9.16% for the unknown group. This indicates that prior campaign outcome is associated with current subscription in the data. It should be interpreted as an association, not a causal effect.

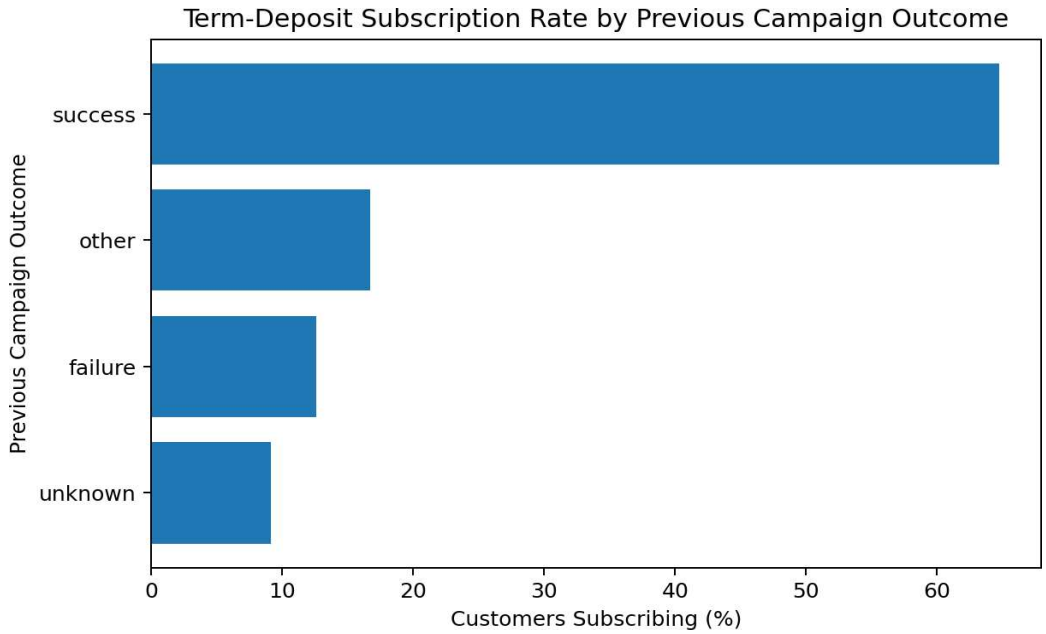


Figure 3. Subscription rate by previous campaign outcome. A recorded previous success is associated with a higher observed rate.

4. Model Evaluation

All four models were trained and evaluated on the same stratified test set. Accuracy measures the overall fraction of correct predictions; precision measures the fraction of predicted subscribers who actually subscribed; recall measures the fraction of actual subscribers detected; F1-score balances precision and recall; ROC-AUC measures ranking discrimination across thresholds.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	F1 rank
Random Forest	0.8954	0.6373	0.2457	0.3547	0.7888	1
Decision Tree	0.8309	0.2982	0.3289	0.3128	0.6132	2
KNN	0.8874	0.5529	0.1975	0.2911	0.7027	3
Logistic Regression	0.8933	0.6632	0.1786	0.2815	0.7717	4

Random Forest achieved the highest accuracy (0.8954), F1-score (0.3547), and ROC-AUC (0.7888). Logistic Regression had the highest precision (0.6632), while Decision Tree had the highest recall (0.3289). The F1 ranking places Random Forest first, Decision Tree second, KNN third, and Logistic Regression fourth.

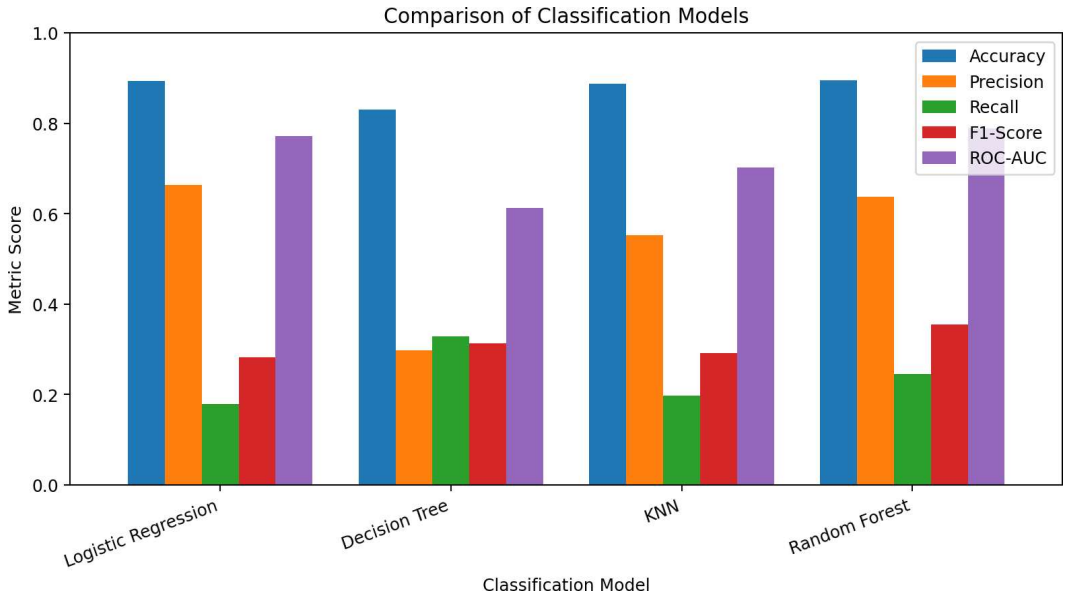


Figure 4. Metric comparison. Random Forest leads in accuracy, F1-score, and ROC-AUC; Decision Tree has the highest recall, and Logistic Regression has the highest precision.

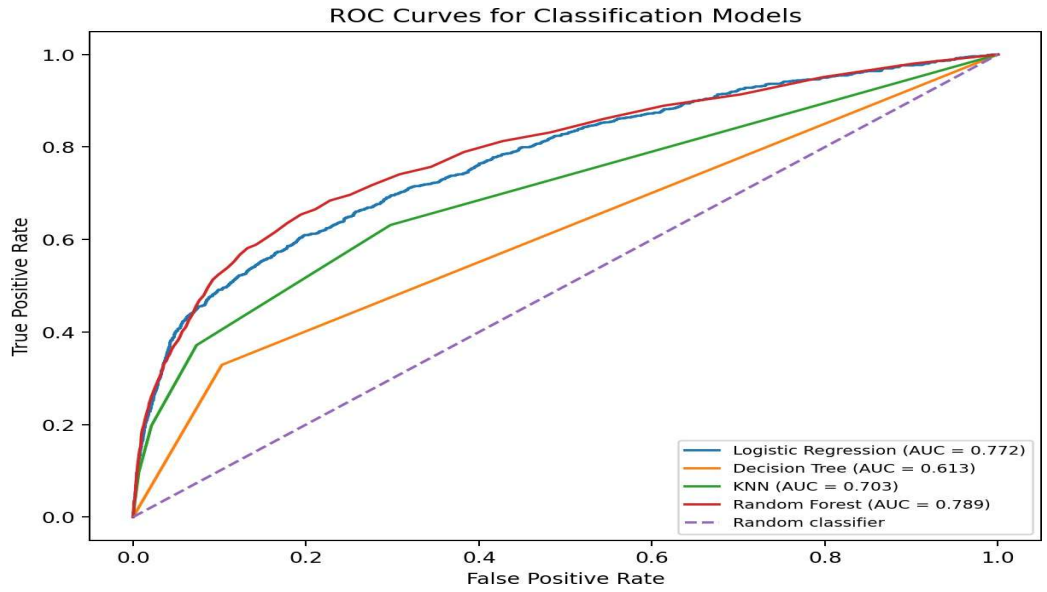


Figure 5. ROC curves. Random Forest has the largest AUC (0.7888), indicating the strongest overall ranking discrimination among the evaluated models.

Random Forest confusion matrix

For Random Forest, the confusion matrix (actual rows: no/yes; predicted columns: no/yes) was [[7,837, 148], [798, 260]]. It correctly identified 260 actual subscribers but missed 798. Its recall of 24.57% means that it identified approximately one-quarter of actual subscribers at the default threshold. This limits its immediate usefulness if the bank's main priority is to reach as many likely subscribers as possible.

5. Conclusion and Recommendations

The experiments show that no single model leads every metric. Random Forest is the recommended initial model for this comparison because it has the best F1-score and ROC-AUC, along with the highest accuracy. However, its recall is low, so its predictions should not be treated as a complete list of likely subscribers.

Recommendation 1: Evaluate decision-threshold tuning on validation data to improve subscriber recall, and compare the resulting precision-recall trade-off before deployment.
Recommendation 2: Consider the business costs of false positives and false negatives, and assess campaign performance using a validation strategy and additional metrics such as the precision-recall curve.

Limitations: The target is imbalanced; unknown categorical values may hide useful distinctions; the models use a single train-test split; and the results depend on the chosen features, preprocessing, and default prediction thresholds. Excluding duration makes the prediction scenario more realistic before a call, but may also reduce predictive performance compared with models that use post-call information.